Fiscal Year:	FY 2024	**Task Last Updated:**	FY 04/15/2024
PI Name:	Marshall-Goebel, Karina Ph.D.		
Project Title:	Exploration Exercise Treadmill Requirements		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	36
Comments:	New affiliation as of spring 2022: NASA Johnson Space Center; Human Physiology, Performance, Protection & Operations (H-3PO) Laboratory New affiliation as of fall 2018: KBR/NASA Johnson Space Center, Cardiovascular and Vision Laboratory, Houston; previously at Massachusetts General Hospital		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	06/27/2022	**End Date:**	09/30/2034
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
Contact Email:	becky.brocato@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Project start date and end date updated due to P.I. change to K. Marshall-Goebel and selection for flight. Original end date: 09/30/2027. Information per T. Finkle/NASA-JSC. (Ed. 8/4/22)		
Key Personnel Changes/Previous PI:	Add Co-I: Millennia Young (NASA Johnson Space Center). Principal Investigator (PI) changed from Marissa Rosenberg to Karina Marshall-Goebel, Ph.D. Per NASA-JSC information (Ed., 8/4/22).		
COI Name (Institution):	Sibonga, Jean (NASA Johnson Space Center) Smith, Scott (NASA Johnson Space Center) Zwart, Sara (NASA Johnson Space Center) Spector, Elizabeth (KBR/NASA Johnson Space Center) Varanoske, Alyssa (KBR/NASA Johnson Space Center) Peters, Brian (KBR/NASA Johnson Space Center) Prejean, Brian (KBR/NASA Johnson Space Center) Young, Millennia (NASA Johnson Space Center)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Exploration class missions including Artemis and beyond are required to have an exercise device that is lightweight and has a small footprint. This has resulted in the development of exercise devices that are motorized and flywheel-based to provide both aerobic and resistive training on one platform. These devices provide a variety of full body resistance exercise options as well as aerobic rowing and cycling, but no treadmill. Because a treadmill has been available for use on the International Space Station (ISS) since 2001, we do not understand what the effect of removing the treadmill will have on muscle performance, aerobic fitness, bone health, sensorimotor performance, or psychological health. Specific Aims: 1) We aim to quantify the effect of cycle ergometry and resistive exercise only, throughout an entire spaceflight mission on muscle performance, aerobic fitness, bone health, sensorimotor performance, or psychological health by removing the treadmill portion of exercise on ISS and replacing it with additional time on the ISS cycle ergometer (CEVIS), as prescribed by the astronaut trainers, while keeping exercise on the Advanced Resistive Exercise Device (ARED) the same. 2) We aim to quantify the effectiveness of the European Enhanced Exploration Exercise Device (E4D) during an entire spaceflight mission on bone, muscle, aerobic, and sensorimotor health and performance by having crewmembers utilize the EED exclusively.
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research. Exploration class missions including Artemis, Gateway, and beyond, are required to have an exercise device that is lightweight and has a small footprint. This has resulted in the development of exercises devices that are motorized and flywheel-based to provide both aerobic and resistive training on one platform. These devices provide a variety of full body resistance exercise options as well as aerobic rowing and cycling, but no treadmill. Without a treadmill, there is no means of ambulating, or reinforcing the motor pattern of walking. This study aims to quantify the effect of no treadmill usage during an entire spaceflight mission on bone, muscle, aerobic, and sensorimotor health and performance.
Research Impact/Earth Benefits:	
Task Progress:	July 2024 Update: This project will be continued under a new Principal Investigator, Alyssa Varanoske. For updated reporting. see https:// (Ed. 7/25/24) In this reporting period, subject recruitment and data collection on active and control subjects has continued. To date, three active subjects (Advanced Resistive Exercise Device/ARED and Cycle Ergometer with Vibration Isolation Stabilization/CEVIS) have been enrolled in the study, and data collection has been completed on one active subject. Eleven control subjects have been enrolled (data share only). Recruitment for one active group (use of ARED and CEVIS only) and the control group (nominal ISS exercise: T2, ARED, and CEVIS) is underway, and informed consent briefings have been presented to crew for two separate missions throughout this reporting period. Recruitment for the second active group (E4D group) is not expected to commence until late 2024. Additionally, changes have been made to the study protocol documents and informed consent to include data sharing of the isometric mid-thigh pull (IMTP) strength test with the Standard Measures study during pre-, in-, and post-flight testing for control subjects. Two familiarization sessions with the IMTP test have also been included in the study protocol.
Bibliography Type:	Description: (Last Updated: 11/19/2025)
Abstracts for Journals and Proceedings	Varanoske AN, Prejean BJ, Strock NC, Conly D, Peters BT, Morant ES, Sibonga JD, Smith SM, Zwart SR, Spector ER, Bishop M, Fincke RS, Young M, Marshall-Goebel K. "Effects of replacing treadmill running with alternative exercise countermeasures during long-duration spaceflight." 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. Abstracts. 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. , Feb-2024